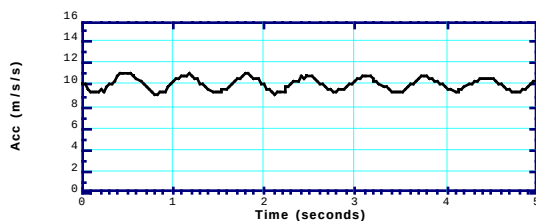


Low-g Accelerometer

(Order Code LGA-BTA or LGA-DIN)

The Low-g Accelerometer can be used for a wide variety of experiments and demonstrations, both inside the lab and outside. Some suggested activities in the laboratory are:

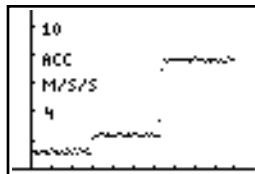
- Measure the acceleration of dynamics carts as they roll down inclines.
- Measure the acceleration of dynamics carts as a force is applied to them. If you have a force sensor, monitor the force at the same time and demonstrate Newton's second law.
- Mount the Low-g Accelerometer on a mass that is hanging from a spring. Start it in simple harmonic motion and monitor the acceleration. If you have a force sensor, measure the force at the same time and study the relationship between acceleration and force.
- Use the Low-g Accelerometer to measure the tilt of an object. Since the Low-g Accelerometer senses the vertical component of gravity, its reading will change as its orientation is changed from horizontal to vertical. You can measure angles to the nearest degree.
- Measure acceleration as you swing the Low-g Accelerometer as a pendulum bob (see graph below).
- Put the Low-g Accelerometer under your belt buckle, and jump up and down. Measure the acceleration as you land, both with your knees flexed and with your knees held more stiff.
- Collect data for an acceleration vs. time and then integrate over time to determine change in speed. Compare with speed determined by another method.



Acceleration of a pendulum bob.

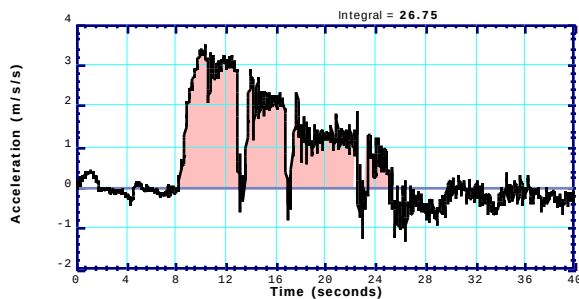
If you have a lab interface that allows you to collect data away from the lab, you might try to measure acceleration:

- on elevators
- on amusement park rides
- on playground apparatus, such as merry-go-rounds
- on remote-control toy cars
- of sky divers or bungee jumpers, etc.
- measure centripetal acceleration on a turntable (see graph on right)
- in a car



Acceleration on a turntable at three different RPMs (Using CBL and a TI-82)

NOTE: This product is to be used for educational purposes only. It is not appropriate for industrial, medical, research, or commercial applications.



*Acceleration of a manual shift automobile from 0 to 60 mph.
(Using Data Logger for Macintosh)*

Using the Accelerometer with a Computer

This sensor can be used with a Macintosh® or PC computer and any of the following lab interfaces: Vernier LabPro™, Universal Lab Interface, or Serial Box Interface. Follow these general procedures to use the Accelerometer with a computer:

1. Connect the Accelerometer to the appropriate port on the interface.
2. Start the data collection software on the computer. If you are using a Power Macintosh or Windows® computer, run the Logger Pro™ software. If you are using an older Macintosh, MS-DOS®, or Windows 3.1 computer, run the Data Logger program.
3. Open an experiment file for the Low-g Accelerometer, and you are ready to collect data.

Using the Accelerometer with TI Graphing Calculators

This sensor can be used with a TI Graphing Calculator and any of the following lab interfaces: LabPro, CBL 2™, or CBL™. Here is the general procedure to follow when using the Accelerometer with a graphing calculator:

1. Load a data-collection program onto your calculator:
 - LabPro or CBL 2: Use the DataMate program. This program can be transferred directly from LabPro or CBL 2 to the TI Graphing Calculator. Use the calculator-to-calculator link cable to connect the two devices. Put the calculator into the Receive mode, and then press the Transfer button on the interface.
 - Original CBL: Use the PHYSICS program. This program is available free on our web site at www.vernier.com. Our programs can also be obtained on disk. (Contact us for more information.) Load the program into a calculator using TI-GGRAPH LINK™.
2. Use the calculator-to-calculator link cable to connect the interface to the TI Graphing Calculator using the I/O ports located on each unit. Be sure to push both plugs in firmly.
3. Connect the Accelerometer to any of the analog ports on the interface. In most cases, Channel 1 is used.
4. Start the data-collection program, and you are ready to collect data.

Specifications Power: 10 mA @ 5 VDC Range: $\pm 49 \text{ m/s/s}$ ($\pm 5 \text{ g}$)
Accuracy: $\pm 0.5 \text{ m/s/s}$ ($\pm 0.05 \text{ g}$) Frequency Response: 0 - 100 Hz

If you purchased the LGA-BTA, you will find a DIN to BTA adapter in the box with your sensor. Use the adapter to connect the sensor to a LabPro, CBL 2 or CBL.

How the Accelerometer Works

The Low-g Accelerometer senses acceleration using an integrated circuit (IC) of a type originally designed to control the release of air bags in an automobile. This IC is micro-machined with very thin “fingers” carved in silicon. These fingers flex when accelerated. They are arranged and connected like the plates of a capacitor. As the fingers flex, the capacitance changes, and a circuit included in the IC monitors the capacitance, converting it into a voltage. An external op-amp circuit amplifies and filters the output from the IC. This voltage can be read by any of the following lab interfaces available from Vernier:

- Vernier LabPro™ (for use with computers or TI Graphing Calculators)
- Texas Instruments CBL 2™ or CBL™ System
- Universal Lab Interface (ULI)
- Serial Box Interface
- MultiPurpose Lab Interface (MPLI)

The Low-g Accelerometer measures acceleration along the line marked by the arrow on the label. Accelerations are normally measured in either meters per second per second (m/s/s) or g’s. One g is the acceleration due to gravity at the earth’s surface, or 9.8 m/s/s. This Accelerometer will measure accelerations in the range of -5 g (-49 m/s/s) to +5 g (+49 m/s/s). This is a range of accelerations which a human body could experience without damage. Many collisions will produce much larger accelerations. In fact, dropping the Accelerometer on a hard surface from even a few centimeters can produce accelerations of a hundred g’s. The Low-g Accelerometer will not be damaged by accelerations up to 1000 g’s.

Note that the Low-g Accelerometer also senses the effect of gravity. We use this to provide an easy way to calibrate the Accelerometer. Also, it allows you to use the Low-g Accelerometer as an “Inclinometer” to measure angles. Its reading will change as its orientation is changed from horizontal to vertical. You can measure angles to the nearest degree.

The Low-g Accelerometer is based on the ADXL05 from Analog Devices. It is designed to measure small accelerations with minimal electronic noise. The noise is typically on the order of 0.5 m/s/s peak to peak. The offset voltage (voltage output at 0 m/s/s) will drift somewhat with temperature. Performing a calibration prior to the experiment will provide the best results. Calibration is very easy with this sensor and is explained on the next page.

Vernier produces two other accelerometers:

- 25-g Accelerometer (order code ACC-BTA, \$91 or ACC-DIN, \$89). Use for collision experiments or for centripetal acceleration with larger accelerations.
- 3-Axis Accelerometer (order code 3D-BTA or 3D-DIN, \$199). Three Low-g units mounted at right angles in one small enclosure. Use for real world collection such as amusement part rides.

Calibration

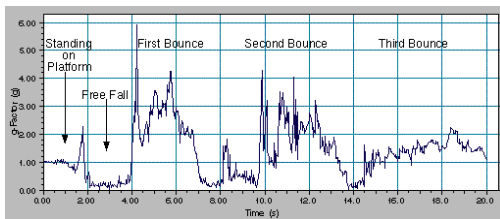
Calibration may be done using the acceleration due to gravity. To calibrate the sensor for measuring acceleration in the horizontal direction, position the Accelerometer with the arrows pointing down for the first calibration point. Define this as -9.8 m/s/s or -1 g . Rotate the Accelerometer so the arrows point up and use the reading for the second calibration point. Define this as $+9.8 \text{ m/s/s}$ or $+1 \text{ g}$. The Accelerometer will then read 0 with no acceleration when held horizontally.

If you want to calibrate for measuring acceleration in the vertical direction, follow the procedure above, but define the first calibration point as 0 m/s/s or 0 g and the second point as 19.6 m/s/s or 2 g .

Using Multiple Accelerometers

You may want to simultaneously measure the signals from two or three accelerometers mounted at right angles. Two Accelerometers mounted at right angles allow you to study acceleration in a plane. This works well, for example, in an automobile. You can mount three Accelerometers at right angles, or better yet use our 3-Axis Accelerometer (3D-DIN), to study acceleration in every direction. This would be good for some amusement park rides and for bungee jumping. Calibrate each Accelerometer for measuring horizontal acceleration. Mount them at right angles. Create a "New Column" using your lab interface program to calculate the square root of the sum of the squares of the accelerations. This value will be equal to 1 g when the combination Accelerometer has no acceleration, and zero when it is in free fall. The orientation of the combination Accelerometer does not matter.

Here is a sample graph made using three Accelerometers mounted at right angles and taken on a bungee jump.



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